Project Export

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1. AnalogEdge - PLL Behavioral Modeling and Verification

Domain: VLSI Slug: analogedge-pll-behavioral-modeling-and-verification-vlsi Created: 4/12/2026

Summary

Develop a mixed-signal PLL project with behavioral models, lock-time analysis, and jitter sensitivity characterization.

Full Content

Project Overview

Case Study: AnalogEdge Create a practical mixed-signal verification project focused on PLL behavior and clock quality constraints.

Key Objectives & Technical Architecture

Phase 1: PLL Block Modeling

- Model PFD, charge pump, loop filter, and VCO behavior.
- Define target frequency and lock conditions.

Phase 2: Dynamic Simulation

- Measure lock time and frequency settling.
- Evaluate phase error under perturbations.

Phase 3: Noise and Jitter Sensitivity

- Run parameter sweeps for loop bandwidth and damping.
- Analyze jitter and stability tradeoffs.

Phase 4: Integration Guidelines

- Generate interface and clocking recommendations.
- Document model limitations and assumptions.

Prerequisites

- Analog/mixed-signal fundamentals
- Control-loop basics

- SPICE/behavioral simulation

Deliverables

1. PLL behavioral model files.
2. Lock/jitter characterization plots.
3. Integration and design recommendation report.

Project Timeline & Deadlines

- Week 1: Block model setup.
- Week 2: Lock behavior simulation.
- Week 3: Jitter and robustness analysis.
- Week 4: Final report and handoff.